## Description

### Image





### Caption

1. Silver bullion. 2. Solid silver conductors.

### The material

If gold is the king of metals, silver is the queen. Silver is a soft, white metal with the highest electrical and thermal conductivities of any metal. It occurs as native silver but most is produced as a by-product of copper, lead and zinc refining.

Silver is valued as a precious metal, use for jewelry, tableware, musical instruments and currency. It has many industrial applications as electrical contacts and conductors, as a catalyst, in photographic film and photo-voltaics, in batteries, in pharmaceuticals, in lead-free solders and in control rods of nuclear reactors. The important industrial uses of silver compete with its desirability as a hedge against inflation, leading to volatile pricing.

### Compositional summary

>99.9Ag

## General properties

| | | | | |
|---|---|---|---|---|
| Density | | 1.05e4 | | kg/m^3 |
| Price | * | 564 | 661 | GBP/kg |
| Date first used | | -4000 | | |

## Mechanical properties

| | | | | |
|---|---|---|---|---|
| Young's modulus | | 69 | 73 | GPa |
| Shear modulus | | 24 | 28 | GPa |
| Bulk modulus | | 100 | 116 | GPa |
| Poisson's ratio | | 0.385 | 0.395 | |
| Yield strength (elastic limit) | * | 190 | 300 | MPa |
| Tensile strength | | 255 | 340 | MPa |
| Compressive strength | * | 190 | 300 | MPa |

| | | | | | |
|---|---|---|---|---|---|
| Elongation | | 1 | - | 2 | % strain |
| Hardness - Vickers | | 90 | - | 110 | HV |
| Fatigue strength at 10^7 cycles | * | 100 | - | 170 | MPa |
| Fracture toughness | * | 58.7 | - | 64.9 | MPa.m^0.5 |
| Mechanical loss coefficient (tan delta) | * | 0.001 | - | 0.002 | |

## Thermal properties

| | | | | | |
|---|---|---|---|---|---|
| Melting point | | 957 | - | 967 | °C |
| Maximum service temperature | * | 96.9 | - | 190 | °C |
| Minimum service temperature | | -273 | | | °C |
| Thermal conductor or insulator? | | Good conductor | | | |
| Thermal conductivity | | 416 | - | 422 | W/m.°C |
| Specific heat capacity | | 230 | - | 240 | J/kg.°C |
| Thermal expansion coefficient | | 19.5 | - | 19.9 | µstrain/°C |

## Electrical properties

| | | | | | |
|---|---|---|---|---|---|
| Electrical conductor or insulator? | | Good conductor | | | |
| Electrical resistivity | | 1.67 | - | 1.81 | µohm.cm |

## Optical properties

| | |
|---|---|
| Transparency | Opaque |

## Critical Materials Risk

| | |
|---|---|
| High critical material risk? | No |

## Durability: water and aqueous solutions

| | |
|---|---|
| Water (fresh) | Excellent |
| Water (salt) | Excellent |
| Soils, acidic (peat) | Excellent |
| Soils, alkaline (clay) | Excellent |
| Wine | Excellent |

## Durability: acids

| | |
|---|---|
| Acetic acid (10%) | Excellent |
| Acetic acid (glacial) | Excellent |
| Citric acid (10%) | Excellent |
| Hydrochloric acid (10%) | Excellent |
| Hydrochloric acid (36%) | Excellent |
| Hydrofluoric acid (40%) | Excellent |
| Nitric acid (10%) | Excellent |

| | |
|---|---|
| Nitric acid (70%) | Excellent |
| Phosphoric acid (10%) | Excellent |
| Phosphoric acid (85%) | Excellent |
| Sulfuric acid (10%) | Unacceptable |
| Sulfuric acid (70%) | Unacceptable |

## Durability: alkalis

| | |
|---|---|
| Sodium hydroxide (10%) | Excellent |
| Sodium hydroxide (60%) | Excellent |

## Durability: fuels, oils and solvents

| | |
|---|---|
| Amyl acetate | Excellent |
| Benzene | Acceptable |
| Carbon tetrachloride | Excellent |
| Chloroform | Excellent |
| Crude oil | Unacceptable |
| Diesel oil | Unacceptable |
| Lubricating oil | Unacceptable |
| Paraffin oil (kerosene) | Unacceptable |
| Petrol (gasoline) | Unacceptable |
| Silicone fluids | Excellent |
| Toluene | Excellent |
| Turpentine | Excellent |
| Vegetable oils (general) | Unacceptable |
| White spirit | Excellent |

## Durability: alcohols, aldehydes, ketones

| | |
|---|---|
| Acetaldehyde | Excellent |
| Acetone | Excellent |
| Ethyl alcohol (ethanol) | Excellent |
| Ethylene glycol | Excellent |
| Formaldehyde (40%) | Excellent |
| Glycerol | Excellent |
| Methyl alcohol (methanol) | Excellent |

## Durability: halogens and gases

| | |
|---|---|
| Chlorine gas (dry) | Excellent |
| Fluorine (gas) | Excellent |
| O2 (oxygen gas) | Excellent |

| | |
|---|---|
| Sulfur dioxide (gas) | Unacceptable |

## Durability: built environments

| | |
|---|---|
| Industrial atmosphere | Excellent |
| Rural atmosphere | Excellent |
| Marine atmosphere | Unacceptable |
| UV radiation (sunlight) | Excellent |

## Durability: flammability

| | |
|---|---|
| Flammability | Non-flammable |

## Durability: thermal environments

| | |
|---|---|
| Tolerance to cryogenic temperatures | Excellent |
| Tolerance up to 150 C (302 F) | Excellent |
| Tolerance up to 250 C (482 F) | Excellent |
| Tolerance up to 450 C (842 F) | Excellent |
| Tolerance up to 850 C (1562 F) | Unacceptable |
| Tolerance above 850 C (1562 F) | Unacceptable |

## Geo-economic data for principal component

| | | | | |
|---|---|---|---|---|
| Annual world production, principal component | 2.4e4 | | | tonne/yr |
| Reserves, principal component | 4.6e5 | | | tonne |

## Primary material production: energy, climate change and water

| | | | | |
|---|---|---|---|---|
| Climate change (CO2-eq) (virgin grade production) | * 452 | - | 498 | kg/kg |
| Embodied energy (virgin grade production) | * 6.87e3 | - | 7.57e3 | MJ/kg |
| Water usage | * 7.25e4 | - | 8.02e4 | l/kg |

## Material processing: energy

| | | | | |
|---|---|---|---|---|
| Metal casting, embodied energy | * 35.6 | - | 39.8 | MJ/kg |
| Metal extrusion, hot, embodied energy | * 7.92 | - | 13.2 | MJ/kg |
| Metal rolling and forging, embodied energy | * 4.43 | - | 7.14 | MJ/kg |
| Wire drawing, embodied energy | * 13.1 | - | 14.5 | MJ/kg |
| Metal powder forming, embodied energy | * 23.8 | - | 27.5 | MJ/kg |
| Chemical vapor deposition (CVD), embodied energy | * 3.7e3 | - | 4.5e3 | MJ/kg |
| Machining, coarse, embodied energy | * 15.7 | - | 17.9 | MJ/kg |
| Machining, fine, embodied energy | * 17.8 | - | 21.6 | MJ/kg |
| Machining, grinding, embodied energy | * 19.9 | - | 25.3 | MJ/kg |
| Machining, laser, CO2, embodied energy | * 0.441 | - | 0.536 | MJ/m |

## Material processing: climate change

| | | | | |
|---|---|---|---|---|
| Metal casting, climate change (CO2-eq) | * 2.44 | - | 2.74 | kg/kg |
| Metal extrusion, hot, climate change (CO2-eq) | * 0.518 | - | 0.866 | kg/kg |
| Metal rolling and forging, climate change (CO2-eq) | * 0.29 | - | 0.468 | kg/kg |
| Wire drawing, climate change (CO2-eq) | * 0.86 | - | 0.95 | kg/kg |
| Metal powder forming, climate change (CO2-eq) | * 1.91 | - | 2.21 | kg/kg |
| Chemical vapor deposition (CVD), climate change (CO2-eq) | * 305 | - | 371 | kg/kg |
| Machining, coarse, climate change (CO2-eq) | * 1.02 | - | 1.18 | kg/kg |
| Machining, fine, climate change (CO2-eq) | * 1.16 | - | 1.42 | kg/kg |
| Machining, grinding, climate change (CO2-eq) | * 1.3 | - | 1.66 | kg/kg |
| Machining, laser, CO2, climate change (CO2-eq) | * 0.0363 | - | 0.0441 | kg/m |

## Recycling and end of life

| | | | | |
|---|---|---|---|---|
| Recycle | ✓ | | | |
| Functional recycle | ✓ | | | |
| Climate change (CO2-eq), recycling | * 13.6 | - | 14.9 | kg/kg |
| Embodied energy, recycling | * 137 | - | 151 | MJ/kg |
| Recycle fraction in current supply | 14 | | | % |
| Combust for energy recovery | ✗ | | | |
| Landfill | ✓ | | | |
| Biodegrade | ✗ | | | |
| A renewable resource? | ✗ | | | |

## Supporting information

### Design guidelines

Silver is non-toxic and has useful anti-bacterial properties.

### Technical notes

Silver is by-product of the electrolytic refining of copper and other metals, notably nickel and zinc. It is extracted from lead by mixing the molten lead with molten zinc, in which the silver preferentially dissolves. The zinc, insoluble in lead, is separated and distilled off, leaving the silver (the Parkes process).

### Typical uses

Electrical contacts, linings for chemical reactor vessels, linings for heavy duty journal bearings, jewelry, table wear, photography, batteries, pharmaceuticals, lead-free solders and control rods of nuclear reactors. Aluminum and rhodium can be substituted for silver in mirrors and other reflecting surfaces. Tantalum can be used in place of silver for surgical plates, pins, and sutures. Stainless steel is an alternate material used widely in the manufacture of table flatware. Nonsilver batteries being developed may replace silver batteries in some applications. Silverless black and white film, xerography, and film with reduced silver content are alternatives to some uses of silver in photography.

## Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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